3 **(a)** A satellite of mass m has a circular orbit of radius r about a planet of mass M. It may be assumed that the planet and the satellite are uniform spheres that are isolated in space.

Show that the linear speed v of the satellite is given by the expression

$$v = \sqrt{\frac{GM}{r}}$$

where G is the gravitational constant.

[1]

(b) Two moons A and B move in circular orbits about a planet, as shown in Fig. 3.1.

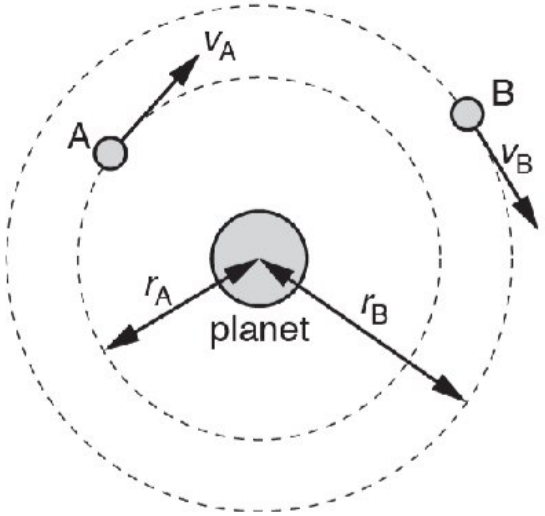


Fig. 3.1

Moon A has an orbital radius r_A of 1.3×10^8 m, linear speed v_A and orbital period T_A.

Moon B has an orbital radius r_B of 2.2×10^{10} m, linear speed v_B and orbital period T_B.

(i) Determine the ratio of $\frac{T_A}{T_B}$.

ratio = ... [3]

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(ii) The planet spins about its own axis with angular speed 1.7×10^{-4} rad s^{-1}. Moon A is always above the same point on the planet's surface.

Determine the orbital period T_B of moon B.

T_B = ... s [2]

(iii) A block is resting on a flat ground located at latitude θ north of the Equator of the planet, as shown in Fig 3.2.

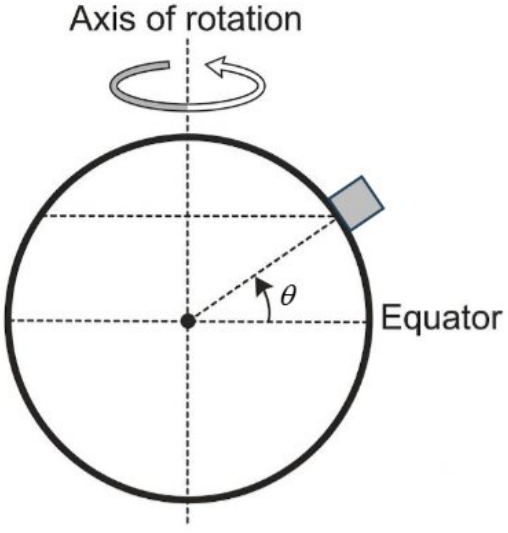


Fig. 3.2

Draw and label all the forces acting on the block in Fig. 3.2. [2]

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